Enterprise Data Governance Framework Design & Implementation for E-Commerce Platform Transformation

By Felice Benita

Project Overview

Objective

Establish a robust, scalable, and compliance-driven Data Governance Framework to ensure:

- High data quality
- Regulatory compliance
- Data ownership clarity
- Trusted analytics & decision-making

Scope

- Customer Data
- Product Data
- Order & Transaction Data
- Vendor/Supplier Data

Executive Summary

- Data fragmentation is limiting business growth and analytics trust
- No clear ownership, weak data quality, and compliance exposure
- Proposed: Enterprise Data Governance Framework
- Outcome: Trusted, compliant, and scalable data foundation

Before vs After Transformation

Before	After
• Siloed data, inconsistent definitions, manual fixes	• Standardized, trusted, and audit-ready data
• Unclear ownership, high error rates	• Governed data with clear ownership

E-Commerce Use Cases

- Customer duplication → incorrect targeting & wasted marketing spend
- Product inconsistency → pricing & catalog errors
- Transaction data issues → fraud detection gaps
- Vendor data mismatch → operational inefficiencies

Business Value

- Expected outcomes: reduced duplicate targeting, fewer data incidents, improved audit readiness, and faster analytics delivery
- Faster decision-making with trusted dashboards
- Improved regulatory compliance readiness

Target Operating Model (Federated)

- Central governance with domain-level ownership
- Chief Data Officer (CDO) & Data Governance Council (executive)
- Data Governance Lead & Domain Data Owners (business)
- Data Stewards & Custodians (operational)

Data Governance
Target Operating Model (TOM)

Link to: 📜 [DATA GOVERNANCE TARGET OPERATING MODEL (TOM)]

Core Governance Capabilities

- Data Quality Management (rules, monitoring, scorecards)
- Metadata & Data Catalog
- Data Lineage & Impact Analysis
- Master Data Management
- Security, Privacy, and Compliance

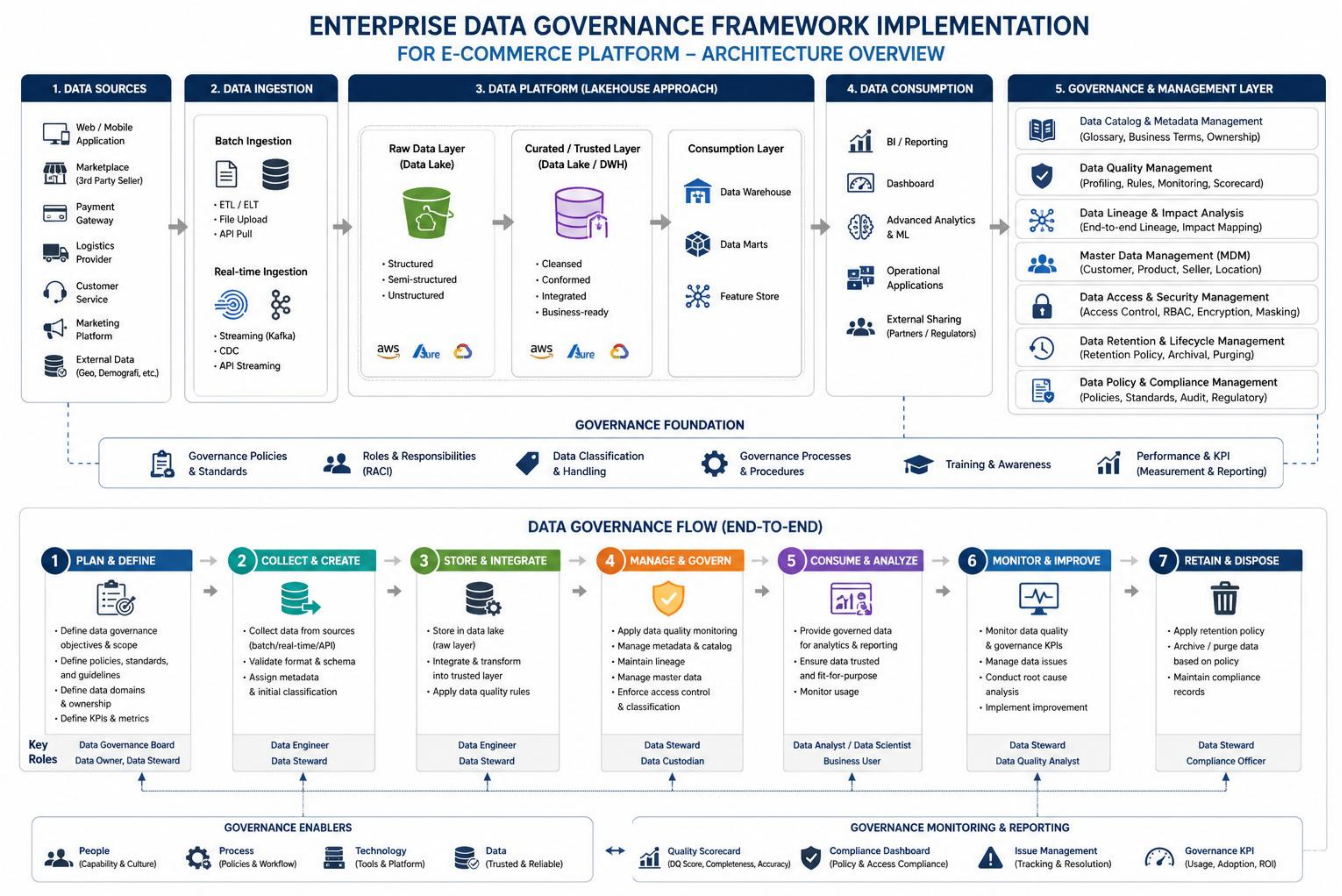
Architecture Overview

- Sources: Web, Mobile, Payment, Logistics
- Ingestion: Batch & Real-time (API/Streaming)
- Storage: Data Lake → Data Warehouse
- Governance Layer across all stages
- Consumption: BI, Analytics, ML
- Governance Services: Catalog, Lineage, DQ Rules, Classification, Access Policies, Audit Logs

End-to-End Governance Flow

- Plan & Define → Collect → Store → Govern
- → Consume → Monitor → Improve → Retain
- Embedded controls across lifecycle
- Continuous feedback loop for improvement

Visual Diagram (Architecture + Governance Flow)



Policy Framework

- Data Ownership & Stewardship mandatory
- Data Classification (Public/Internal/Confidential/Restricted)

  → PII = Restricted
- RBAC-based access control
- Retention & lifecycle rules enforced

Gap & Maturity Assessment

- Current maturity: Level 1–2 (ad hoc, inconsistent)
- Target maturity: Level 4 (managed, standardized)
- Key gaps: ownership, metadata, quality controls
- Priority: high-impact domains first (customer, product)

→ Illustrative maturity assessment aligned to DAMA-DMBOK / CMMI-style governance maturity levels.

Implementation Roadmap

- Phase 1: Foundation (roles, governance setup)
- Phase 2: Policies & standards
- Phase 3: Technology enablement
- Phase 4: Monitoring & continuous improvement

Data Governance Policy

Link to: 📜 DATA GOVERNANCE POLICY

KPIs & Success Metrics

- Data Quality Score >95%
- 100% critical data with assigned owner
- Reduction in incidents & data issues
- Compliance audit success rate

Closing

- Data Governance is a business enabler, not just control
- Critical for scaling e-commerce operations
- Foundation for AI, analytics, and personalization

THANK YOU